

■ DEEPDRISHTI

Agentic AI Flow for Intelligent Underwater Vision

SDG 14 • LIFE BELOW WATER

An autonomous AI system that transforms challenging underwater visual data into enhanced, detected, depth-aware and actionable intelligence.



1. Problem Statement

Underwater environments are difficult to observe because of murky water, low-light conditions, color distortion and limited visibility. Marine researchers, underwater robotics teams and defense-oriented operators often depend on manual review of large volumes of underwater imagery and video. This process is slow, inconsistent and lacks contextual information such as object identity, relative depth and potential risk. Existing solutions often focus on only one task—enhancement, detection or depth estimation—without intelligently combining the outputs.

2. Proposed Solution

DeepDrishti is an Agentic AI workflow that acts as an intelligent visual-analysis layer for underwater environments. It enhances poor-quality underwater imagery, detects relevant objects, estimates relative depth and reasons over the combined results to generate actionable intelligence.

3. Innovation & Unique Value Proposition

DeepDrishti is not simply an underwater image-enhancement model or an object detector. Its core innovation is the orchestration of multiple AI capabilities into a single reasoning workflow: **Enhance** → **Detect** → **Estimate Depth** → **Reason** → **Recommend**. This moves from raw underwater pixels to contextual intelligence.

4. Why Agentic AI?

A conventional AI pipeline produces isolated outputs. DeepDrishti uses an Agentic AI architecture where specialized models are coordinated through a structured workflow. LangGraph manages the multi-step process, while Groq-powered reasoning interprets the combined results and supports future decision logic.

5. Agentic AI Workflow & Technology Stack

Stage	Technology	Function
■ Input	Underwater Image / Video	Receives visual data from cameras, footage or future AUV systems.
■ Enhance	FUNIE-GAN	Improves visibility, color balance and quality in murky or low-light scenes.
■ Detect	YOLOv8	Detects fish, mines, submarines, structures, trash and humans.
■ Depth	MiDaS	Provides relative depth information and spatial context.
■ Orchestrate	LangGraph	Coordinates the multi-step Agentic AI workflow.
■ Reason	Groq LLM	Interprets model outputs and generates actionable intelligence.

6. Target Users & Applications

- **Marine Researchers:** Automated analysis of underwater imagery and biodiversity monitoring.
- **Conservation Teams:** Detection of marine life, underwater waste and environmental conditions.
- **AUV & Robotics Developers:** Visual intelligence for autonomous underwater vehicles.
- **Defense & Security:** Potential support for underwater object and mine-detection scenarios.
- **Academic Institutions:** A platform for experimenting with advanced Computer Vision and Agentic AI.

7. SDG 14 Alignment — Life Below Water

DeepDrishti directly supports SDG 14 by improving the ability to observe and understand underwater ecosystems. Better visual intelligence can help researchers monitor marine life, identify underwater waste, support conservation activities and improve the efficiency of ocean exploration. By converting underwater data into faster and more useful insights, the platform supports responsible understanding and protection of marine environments.

8. Expected Impact

- Reduce manual underwater video-analysis time.
- Improve visibility and object detection in difficult underwater conditions.
- Combine multiple AI outputs to provide contextual intelligence.
- Support marine research, conservation and responsible ocean exploration.
- Provide a scalable foundation for future real-time underwater AI systems.

9. Evaluation Metrics

- FPS / latency for processing speed
- YOLOv8 mAP, precision and recall for detection quality
- Depth consistency for relative depth estimation
- Agentic workflow success rate
- Reduction in manual analysis time

10. Future Scope

Future development can include real-time video streaming, edge deployment on underwater robotics, integration with sonar and other sensor modalities, improved distance estimation, adaptive agent decision-making and deployment directly on AUV platforms. The architecture can also evolve into a multimodal underwater intelligence system combining visual, depth and sensor data.

11. Conclusion

DeepDrishti Agentic AI Flow presents a practical approach to underwater intelligence by combining advanced Computer Vision with Agentic AI orchestration. Through the coordinated use of FUNIE-GAN, YOLOv8, MiDaS, LangGraph and Groq, the system transforms challenging underwater visual data into structured and actionable intelligence. Its alignment with SDG 14 makes it both a technical innovation and a solution with potential environmental and societal impact.

12. Project Links

GitHub Repository	github.com/yuvraj648/DeepDrishti-Agent
Live Streamlit Demo	deepdrishti-agent-6pu9tfun6uoakqkgevayee.streamlit.app
Documentation	Available through the GitHub repository README and project documentation.